

Instruction emphasis can affect suppression (model-dependent). Changing the wording of the suppression prompt to protect the secret greatly affects leakage, as shown in [Appendix H](#). On Opus 4.6, the most emphatic instruction yields the strongest suppression and the most leakage, while casual phrasings induce almost no suppression and drop the attack to chance. GPT-5.4 behaves similarly. In contrast, Gemini 3.1 Pro suppresses almost completely under every wording, so it leaks strongly without any emphasis trend. The data supports suppression as the underlying mechanism and contradicts intuition: instructing a model more forcefully to protect a value makes that value easier to recover.

More training correlates with stronger capabilities and more leakage. This is supported by our OLMo-3-32B-Think results ([Figure 3](#), right), showing that post-training stages (especially SFT and RLVR) each increase leakage in aggregate. Across the post-training stages of OLMo-3-32B-Think, channel entropy falls from 2.91 bits at Base to 1.37 bits after RLVR. SFT contributes -0.98 bits and RLVR a further -0.52 bits, both significant under a paired per-prompt test, while DPO is flat.

5 Discussion and Limitations

Fundamental tension. The value proposition of personal agents requires in-context access to private data (emails, documents, credentials), but our results show that such access implies vulnerability. This is an architectural challenge: as long as secrets share the context window with the generation process, the output distribution carries information about those secrets.

Toward mitigation. Alignment training, output filters, and privacy instructions are demonstrably insufficient. In the passive setting, the model correctly refuses direct extraction yet still leaks. In the active setting, any specific defense (e.g. keyword filters, output sanitization) becomes an additional constraint for the adversary to optimize against, as demonstrated by our RL and rejection-sampling attacks. The mechanism suggests opportunities to reduce leakage. A defense would need the model’s output distribution to be approximately invariant to the protected value. This is predominantly a training-time calibration independent of filtering. For example, when the secret space is small and well-defined, we can force the emitted-secret distribution to be uniform. For arbitrary natural-language secrets, we do not know the set of all strings to which generation needs to be invariant, so it is unclear how to train a model to generate text in this way. Intuitively, one may hope that training on “enough” secrets generalizes, but this requires further exploration.

Centralized models amplify individual risk. A small number of frontier models generate responses for millions of users, so a single adaptive decoder trained once against the public API generalizes across that model’s entire user population. The adversary pays the estimation cost once and amortizes it over every victim ([Table 9](#)), which makes the leakage we measure a population-level risk that compounds with deployment scale.

Limitations. Our experiments target two predicate families—digit values and binary attributes—on a fixed set of models. We leave generalization to other predicate types, models, and deployment configurations open. Open-vocabulary, free-text secrets are a natural next step that we do not attempt here, since the attribute-inference reduction of [Section 2](#) assumes a finite value set, and an unbounded one breaks the $\lceil \log_2 m \rceil$ construction. All eight evaluated models are proprietary and reachable only through their APIs, so we evaluate the black-box case alone ([Section 2](#)). The attack also assumes the adversary can identify the target model and pay a one-time estimation cost against it. When the generating model is unknown, the adversary may need to do more work, but they can likely still succeed: we observe that decoders transfer partially across models ([Figure 7](#)), suggesting an advanced adversary may be able to train against an ensemble of models. We do not evaluate the stronger threat model where the adversary cannot query the model at will. The digit and SSN experiments use uniformly random secrets, whereas real-world secrets carry structure, such as date-formatted PINs or geographically constrained SSNs, that could either help or hinder reconstruction. The memory-predicate evaluation covers a single user profile from CIMemories. Our capability correlation uses Arena Elo as a proxy at $n=8$ from an unversioned leaderboard snapshot. Finally, we train decoders per model, and we do not study how deployment-side mitigations such as output perturbation or response-length normalization would degrade accuracy.

6 Related Work

To our knowledge, this is the first work to study inadvertent context leakage as an independent phenomenon. We position our findings with respect to prior work as follows.

Prompt injection. Prompt injection attacks manipulate LLM behavior by inserting malicious instructions into model inputs (Perez and Ribeiro, 2022). Greshake et al. (2023) extended this to *indirect* prompt injection via adversarial content in external sources, and Christodorescu et al. (2026) systematized overt exfiltration channels in agentic systems. Unlike these works, which focus on hijacking model behavior or exfiltrating data through explicit channels, our threat model considers an adversary who extracts private information through *covert* semantic channels.

Training-time and multi-agent threats. Data poisoning attacks manipulate training data to induce targeted misbehavior (Wan et al., 2023; Bagdasaryan and Shmatikov, 2022), backdoors can persist through safety training (Hubinger et al., 2024; Chaudhari et al., 2025), and steganographic encoders can be embedded via fine-tuning (Meier et al., 2025) or multi-agent collusion (Motwani et al., 2024; Cloud et al., 2025). All of these require training-time access or coordination between multiple models; we study what can be extracted from a single, unmodified model at inference time via its natural outputs.

Language model inversion. Model inversion attacks reconstruct private inputs from model outputs. Morris et al. (2024) showed model outputs can be inverted to recover input prompts; Zhang et al. (2022) demonstrated similar attacks on classifiers; and Song and Raghunathan (2020) and Li et al. (2023) showed that sentence embeddings reveal sensitive attributes. These attacks typically require access to model logits or embeddings; our adversary observes only the text output (i.e., the information exposed by frontier APIs).

Privacy in language models. Several frameworks address LLM privacy: contextual-integrity-based violation detection (Li et al., 2025; Shvartzshnaider and Duddu, 2025), privacy norm evaluation for agents (Shao et al., 2024), fine-tuning for contextual privacy (Xiao et al., 2024), and user disclosure risk studies (Zhang et al., 2024). These works focus on preventing *voluntary* disclosure; we study settings where the model discloses nothing explicitly, yet information leaks through the statistical properties of benign outputs.

Privacy in in-context learning. A parallel line studies privacy for in-context learning itself. Closest to us, Choi et al. (2025) audit privacy-preserving ICL methods: they insert identifiable canary tokens into prompt exemplars, craft queries to surface them, and compare the resulting worst-case leakage against a mechanism’s theoretical privacy budget. Both papers ask what escapes a model’s context, and the two approaches are complementary. In their threat model, a model that refuses the extraction request has protected the data, and that refusal is where our work begins.

Latent knowledge. Treutlein et al. (2024) showed that LLMs infer latent structure from disparate training documents, Betley et al. (2025) demonstrated behavioral self-awareness, and Cywiński et al. (2025) studied eliciting knowledge that models possess but refuse to verbalize. These works suggest that LLMs maintain latent representations that could surface in output distributions even when the model refuses to state them explicitly, motivating our study of in-context secret leakage.

7 Conclusion

We have shown that in-context secrets are not safe: even when a language model correctly refuses to reveal a secret, its benign outputs carry nonzero information about the secret’s value. The phenomenon extends to natural language, where semantic predicates about user memories are inferrable from routine task outputs. This passive leakage enables active exploitation: an RL-trained adversary extracts full Social Security Numbers from production-style agents powered by two frontier models (Claude Opus 4.6 and Gemini 3.1 Pro). Our findings indicate that defending in-context secrets requires more than alignment training and output filtering; information-theoretic guarantees on the generation process may be necessary.

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